List-Recoverable Subcodes of AG Codes

Theorem 1.1. *Let $\varepsilon > 0$ and $\ell \in \mathbb{N}^+$. Let $q = O(1/\varepsilon^2)$ be a large enough even power. Choose $R \in (0,1)$. There is some $m_{\min} = O(\ell \log_q(\ell/\varepsilon)/\varepsilon^2)$ such that the following holds. For all large enough n of the form $q^{e/2}(\sqrt{q}-1)$, where $e > 0$ is an integer, and all $m \geq m_{\min}$, there is a deterministic construction of an $\mathbb{F}_q$-linear code $C : \mathbb{F}_{q^m}^{Rn} \to \mathbb{F}_{q^m}^n$ over the alphabet $\mathbb{F}_{q^m}$ of rate at least R and relative distance $1-R-O(\varepsilon)$ that is $(1-R-\varepsilon, \ell, L)$ list-recoverable in time $\mathrm{poly}(L,n)$, returning a list of size at most $L := \exp_q(\mathrm{poly}(\ell/\varepsilon))$ that is contained in a subspace over $\mathbb{F}_q$ of dimension at most $\mathrm{poly}(\ell/\varepsilon)$.*

Moreover, we can reduce the size of the alphabet in our theorem 1.1 by concatenation. This concatenation exactly follows the same proof techniques of Hemenway, Ron-Zewi, and Wootters [HRZW19] in their Appendix A.

Theorem 1.2. *There are constants c, c_0 so that the following holds. Choose $\varepsilon > 0$ and a positive integer ℓ. Suppose that $q \geq \ell^{c/\varepsilon}$ is an even power of a prime. Let $n_0 = \exp(\exp_q(c_0\ell/\varepsilon))$. Then for all $n \geq n_0$, there is a deterministic polynomial-time construction of an $\mathbb{F}_q$-linear code $C : \mathbb{F}_q^{(1-\varepsilon)n} \to \mathbb{F}_q^n$ of rate $1-\varepsilon$ and relative distance $\Omega\left(\varepsilon^2\right)$ which is $\left(\Omega\left(\varepsilon^2\right), \ell, L\right)$ list-recoverable for $L = \exp_q\left(\mathrm{poly}\left(\frac{\exp_q(c\ell/\varepsilon)}{\varepsilon}\right)\right)$.*

Proof sketch. The proof follows essentially the same structure as the proof of [HRZW19, Theorem A.1] given by Hemenway, Ron-Zewi, and Wootters. Their proof was based on a concatenation framework presented in [HRZW19, Lemma 7.2], which combines the inner code from [HRZW19, Corollary 4.4] with the outer code from [HRZW19, Theorem A.7]. The key difference in our proof is that we provide an improved outer code (see Theorem 1.1), which improves the dimension of the affine space containing the output list to $\mathrm{poly}(\ell/\varepsilon)$. Consequently, the output list size L of our final code, obtained by concatenating the inner codes from [HRZW19, Corollary 4.4] with the outer codes from Theorem 1.1, is bounded by:

$$L \leq \exp_q\left(\mathrm{poly}\left(\frac{\exp_q(c\ell/\varepsilon)}{\varepsilon}\right)\right).$$

As a side remark, the block length of our outer code must be at least double-exponential in ℓ/ε, specifically at least $n_0 = \exp(\exp_q(c_0\ell/\varepsilon))$, whereas in [HRZW19], it was only single-exponential. This requirement arises because the block length of our outer code must satisfy a lower bound $k_{\min}$, imposed by the proof of Theorem 1.6 below. However, this does not introduce additional difficulties in the proof, and n_0 remains a constant when q, ℓ, and ε are treated as constants. □

The rest of this section is devoted to the proof of Theorem 1.1.

1.1 BTT Evasive Subspaces

Guo and Ron-Zewi [GRZ21] introduced the notion of BTT matrices and BTT subspaces, which are matrices and subspaces exhibiting nice periodic behaviors. They further define BTT-evasive subspaces, which are subspaces of low codimension that have a small intersection with all BTT-subspaces. We now formally define these objects.

Definition 1.3 (BTT matrix and subspace)**.** A (k, m, r)*-block-triangular-Toeplitz (*BTT*) matrix over* $\mathbb{F}_q$ is a $(km) \times (kr)$ matrix M over $\mathbb{F}_q$ such that $M = (M_{i,j})_{i,j\in[k]}$, as a $(k \times k)$-block matrix with $m \times r$ blocks $M_{i,j}$, satisfies the following conditions:

1. M is block-lower-triangular, i.e., $M_{i,j} = 0$ for $i, j \in [k]$ with $i < j$.
2. M is block-Toeplitz, i.e., $M_{i,j} = M_{i',j'}$ for $i, j, i', j' \in [k]$ with $i - j = i' - j'$.
3. M has maximal rank.

For a (k, m, r)-BTT matrix M, the image of $\mathbb{F}_q^{kr}$ in $\mathbb{F}_q^{km}$ under the linear map $u \mapsto Mu$ is called a (k, m, r)-BTT *subspace*.

We also need the following relaxations of BTT matrices and subspaces.

Definition 1.4. A matrix M is said to be a (k, m, r)*-periodic matrix* if it satisfies all the conditions of BTT matrices except that the second condition is relaxed to the condition that $M_{i,i} = M_{i',i'}$ for $i, i' \in [k]$. For a (k, m, r)-periodic matrix M, the image of $\mathbb{F}_q^{kr}$ in $\mathbb{F}_q^{km}$ under the linear map $u \mapsto Mu$ is called a (k, m, r)*-periodic subspace*.

$$\begin{pmatrix} M_1 & 0 & 0 & \cdots & 0 \\ M_2 & M_1 & 0 & \cdots & 0 \\ M_3 & M_2 & M_1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ M_k & M_{k-1} & M_{k-2} & \cdots & M_1 \end{pmatrix}$$

Figure 1: A (k, m, r)-BTT matrix consists of blocks M_i, where each M_i is an $m \times r$ matrix, and M_1 (and hence M) has maximal rank.

Definition 1.5 (BTT/periodic evasive subspace)**.** A subspace $W \subseteq \mathbb{F}_q^{km}$ is said to be a (k, m, r, s)-BTT *(resp. periodic) evasive subspace* if for every (k, m, r)-BTT (resp. periodic) subspace $V \subseteq \mathbb{F}_q^{km}$, it holds that $\dim(V \cap W) \leq s$.

The main result of this subsection is the following theorem, which gives an explicit construction of BTT evasive subspaces.

Theorem 1.6. *Let $C > 0$ be a large enough constant, and let $k_{\min} = \exp(\mathrm{poly}(r/\varepsilon))$ be sufficiently large. For $\varepsilon > 0$, prime power q, and $m, r, k \in \mathbb{N}^+$ with $m \geq Cr/\varepsilon \cdot \lceil \log_q(r/\varepsilon) \rceil$ and $k \geq k_{\min}$, there exists a $(k, m, r, \mathrm{poly}(r/\varepsilon))$-*BTT *evasive subspace $W \subseteq \mathbb{F}_q^{km}$ of codimension at most εkm. Moreover, a basis for W can be found in time* $\mathrm{poly}(q, k, m)$.

To prove Theorem 1.6, we need the following lemmas.

Lemma 1.7 ([GRZ21, Lemma II.4])**.** *For $\varepsilon > 0$, $k, m, r \in \mathbb{N}^+$ with $r < \frac{\varepsilon m}{2}$, $s = \frac{2r}{\varepsilon}$, and prime power q, there exists a (k, m, r, s)-*BTT *evasive subspace $W \subseteq \mathbb{F}_q^{km}$ of codimension at most εkm.*

Lemma 1.8 ([GRZ21, Lemma II.7])**.** *Suppose that W is a (k, m, r, s)-*BTT *evasive subspace, and W' is a (k', km, s, s')-periodic evasive subspace. Then $W^{k'} \cap W'$ is a $(k'k, m, r, s')$-*BTT *evasive subspace.*

Lemma 1.9 ([GRZ21, Corollary III.1])**.** *There exists an absolute constant $c > 1$ such that for $\varepsilon > 0$, $k, m, r \in \mathbb{N}^+$ with $r < \frac{\varepsilon m}{4}$, $s = \frac{2r^2}{\varepsilon}$, and prime power q satisfying $q^m \geq \max\left\{k^{c \cdot r/\varepsilon}, \left(\frac{2r}{\varepsilon}\right)^{2r/\varepsilon}\right\}$, there exists a (k, m, r, s)-periodic evasive subspace $W \subseteq \mathbb{F}_q^{km}$ of codimension at most εkm. Moreover, a basis for W can be found in time $\mathrm{poly}(q, k, m)$.*

Proof of Theorem 1.6. Fix $i = 2$. Let $\varepsilon_0 = \varepsilon/(i+1)$. Let $s_i = \frac{2r^2}{\varepsilon_0}$ and $s_j = \frac{2s_{j+1}^2}{\varepsilon_0}$ for $j = i-1, i-2, \ldots, 0$. Note that $s_0, \ldots, s_i \leq \mathrm{poly}(r/\varepsilon)$. Let $k_0 = k$ and $k_j = (r/\varepsilon)^{c_0} \log k_{j-1}$ for $j = 1, 2, \ldots, i$, where $c_0 > 0$ is a large enough constant. Let c be the constant in Lemma 1.9.

We claim that for $j \in \{0, 1, \ldots, i\}$, there exists a (k_j, m, r, s_j)-BTT evasive subspace $W_j \subseteq \mathbb{F}_q^{k_j m}$ of codimension at most $(i-j+1) \cdot \varepsilon_0 k_j m$ whose basis can be found in time $\mathrm{poly}(q, k, m)$. We prove this claim by inducting on j.

First consider the case where $j = i$. We further divide this into two sub-cases:

Sub-Case 1: $k_i^2 m^2 \log q \leq \log k$. In this case, there exists a (k_i, m, r, s_i)-BTT evasive subspace of codimension at most $\varepsilon_0 k_i m$ by Lemma 1.7. By brute-force search, its basis can be found in time $q^{O((k_i m)^2)} = 2^{O(k_i^2 m^2 \log q)} \leq \exp(O(\log k)) = \mathrm{poly}(k)$.

Sub-Case 2: $k_i^2 m^2 \log q > \log k$. In this case, by the choice of parameters, we have

$$r < \frac{\varepsilon_0 m}{4} \quad \text{and} \quad q^m \geq \max\left\{k_i^{c \cdot r/\varepsilon_0}, \left(\frac{2r}{\varepsilon_0}\right)^{2r/\varepsilon_0}\right\}.$$

To see $q^m \geq k_i^{c \cdot r/\varepsilon_0}$, note that it is equivalent to that $m \log q \geq c \cdot r/\varepsilon_0 \cdot \log k_i$. We have $k_i \leq \mathrm{poly}(r/\varepsilon) \log^{\circ i} k \leq \mathrm{poly}(r/\varepsilon)(\log k)^{o(1)}$, where $\log^{\circ i} k$ denotes the i-th iterated logarithm of k. As $k_i^2 m^2 \log q > \log k$, we have $m \log q \geq (m^2 \log q)^{1/2} \geq \left(\frac{\log k}{k_i^2}\right)^{1/2} = \frac{(\log k)^{1/2 - o(1)}}{\mathrm{poly}(r/\varepsilon)}$. It is at least $c \cdot r/\varepsilon_0 \cdot \log k_i$ since $k_i = (\log k)^{o(1)}$, $k \geq k_{\min}$, and $k_{\min} = \exp(\mathrm{poly}(r/\varepsilon))$ is sufficiently large.

Therefore, by Lemma 1.9, there exists a (k_i, m, r, s_i)-periodic evasive subspace of codimension at most $\varepsilon_0 k_i m$ whose basis can be found in time $\mathrm{poly}(q, k, m)$. And it is automatically a (k_i, m, r, s_i)-BTT evasive subspace by definition.

So for both sub-cases, the claim for $j = i$ is proved.

Now assume that $j < i$ and the claim is proved for $j+1 \in [i]$, and we prove the claim for j. Let W_{j+1} be a (k_{j+1}, m, r, s_{j+1})-BTT evasive subspace of codimension at most $(i-j) \cdot \varepsilon_0 k_{j+1} m$ whose basis can be found in time $\mathrm{poly}(q, k, m)$. It exists by the induction hypothesis.

By the choice of parameters, we have

$$s_{j+1} < \frac{\varepsilon_0 k_{j+1} m}{4} \quad \text{and} \quad q^{k_{j+1} m} \geq \max\left\{\left(\frac{k_j}{k_{j+1}}\right)^{c s_{j+1}/\varepsilon_0}, \left(\frac{2s_{j+1}}{\varepsilon_0}\right)^{2s_{j+1}/\varepsilon_0}\right\}.$$

So by Lemma 1.9, there exists an explicit $(k_j/k_{j+1}, k_{j+1} m, s_{j+1}, s_j)$-periodic evasive subspace W' of codimension at most $\varepsilon_0 k_j m$ whose basis can be found in time $\mathrm{poly}(q, k, m)$. Let $W_j = W_{j+1}^{k_j/k_{j+1}} \cap W'$. By Lemma 1.8, W_j is a (k_j, m, r, s_j)-BTT evasive subspace. Its basis can be found in time $\mathrm{poly}(q, k, m)$ since the same holds for W_{j+1} and W'. It is codimension is at most $k_j/k_{j+1} \cdot (i-j) \cdot \varepsilon_0 k_{j+1} m + \varepsilon_0 k_j m = (i-j+1) \cdot \varepsilon_0 k_j m$, as desired. This proves the claim.

Finally, setting $j = 0$ proves the theorem. □

Remark. As observed in [HRZW19, Remark A.3], one may also choose $m_{\min} = O(r/\varepsilon^2)$ in Theorem 1.6, but the field size q would have to depend on r. Such a dependence would cause trouble in code concatenation in the proof of Theorem 1.2. So, we follow [HRZW19] and choose $m \geq Cr/\varepsilon \cdot \lceil \log_q(r/\varepsilon) \rceil$.

1.2 Algebraic-Geometric Codes

Algebraic-geometric (AG) codes are natural generalizations of Reed–Solomon codes, with one advantage being that they can be defined over a small finite field while still having an almost optimal trade-off between the rate and the distance. A detailed treatment of AG codes can be found in, e.g., [Sti09].

Function fields, discrete valuations, and places. Every AG code is associated with a function field (of dimension one) over $\mathbb{F}_q$. Given such a function field F, a (normalized) *discrete valuation* of F is a map $v : F \to \mathbb{Z} \cup \{\infty\}$ such that

- $v(a) = \infty$ iff $a = 0$.
- $v(ab) = v(a) + v(b)$ for $a, b \in F$.
- $v(a + b) \geq \min\{v(a), v(b)\}$ for $a, b \in F$.
- $v(F^\times) = \mathbb{Z}$.

For a discrete valuation v of F, we associate a pair $P = (\mathcal{O}_v, \mathfrak{m}_v)$ where $\mathcal{O}_v$ is the ring $\{a \in F : v(a) \geq 0\}$ and $\mathfrak{m}_v$ is the ideal $\{a \in \mathcal{O}_v : v(a) > 0\}$ of $\mathcal{O}_v$. Call P a *place* of F and define

$\kappa_P := \mathcal{O}_v/m_v$. We say P is *rational* if $\dim_{\mathbb{F}_q} \kappa_P = 1$. For a rational place $P = (\mathcal{O}_v, \mathfrak{m}_v)$, the *evaluation* of $f \in \mathcal{O}_v$ at P is $f(P) := f + \mathfrak{m}_v \in \kappa_P$, which we view as an element of $\mathbb{F}_q$ via the unique $\mathbb{F}_q$-linear isomorphism $\kappa_P \cong \mathbb{F}_q$.

It can be shown that a discrete valuation v of F uniquely corresponds to a place P of F, and we write $v = v_P$. Denote by $\mathbb{P}(F)$ the set of all places of F.

Divisors and Riemann–Roch spaces. A *divisor* of F is a formal sum $D = \sum_P n_P P$ of finitely many places P of F with integer coefficients n_P. Its degree is $\deg(D) := \sum_P n_P [\kappa_P : \mathbb{F}_q]$. The set of all divisors of F forms an abelian group $\mathrm{Div}(F)$ such that $\deg(D + D') = \deg(D) + \deg(D')$. We say a divisor D is *rational* if every place of F appearing in D is rational.

For a divisor $D = \sum n_P P$ of F, the *Riemann–Roch space* associated with D is

$$L(D) := \{f \in F^\times : v_P(f) + n_P \geq 0 \text{ for all } P \in \mathbb{P}(F)\} \cup \{0\}$$

which is a finite-dimensional vector space over $\mathbb{F}_q$. The integer n_P is taken to be zero if P does not appear in D. It is known that $L(D) = 0$ if $\deg(D) < 0$.

It is also known that $\dim_{\mathbb{F}_q} L(D) \in [d + 1 - g, d + 1]$, where $g \geq 0$ is an invariant of the function field F called the *genus* of F. We state the lower bound formally:

Theorem 1.10 (Riemann's inequality)**.** $\dim_{\mathbb{F}_q}(D) \geq \deg(D) - g + 1$*, where g is the genus of F.*

Algebraic-geometric codes. Let D be a divisor of F and let $S = \{P_1, P_2, \ldots, P_n\}$ be a set of n distinct rational places of F not appearing in D. Define the *algebraic-geometric (*AG*) code*

$$C(S, D) := \{(f(P_1), f(P_2), \ldots, f(P_n)) : f \in L(D)\} \subseteq \mathbb{F}_q^n, \tag{2}$$

which is an $\mathbb{F}_q$-linear code of block length n.

1.3 The Garcia–Stichtenoth Tower

The AG code we require is derived from a tower of function fields known as the Garcia-–Stichtenoth tower, introduced by Garcia and Stichtenoth in [GS96].

Definition 1.11 (Garcia–Stichtenoth tower [GS96])**.** Let $r > 1$ be a prime power and $q = r^2$. For $i = 1, 2, \ldots,$ let $K_i = \mathbb{F}_q(x_1, x_2, \ldots, x_i)$, where x_1 is transcendental over $\mathbb{F}_q$ and x_i satisfies the following recursive equation for $i > 1$.

$$x_i^r + x_i = \frac{x_{i-1}^r}{x_{i-1}^{r-1} + 1}. \tag{3}$$

The *Garcia–Stichtenoth tower* over $\mathbb{F}_q$ is the infinite tower of function fields $K_1 \subseteq K_2 \subseteq \cdots$.

Let $e > 0$ be an integer. The genus $g(K_e)$ of K_e is known to be at most r^e. The field K_e has at least $r^e(r-1) + 1$ rational places. One of them is the point "at the infinity," which we denote by P_∞. It is known that $v_{P_\infty}(x_e) = -1$. In addition, define S_e to be the set of all tuples $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_e) \in \mathbb{F}_q^e$ such that $\alpha_1^r + \alpha_1 \neq 0$ and $\alpha_i^r + \alpha_i = \frac{\alpha_{i-1}^r}{\alpha_{i-1}^{r-1}+1}$ for $i = 2, 3, \ldots, e$. For each $\alpha \in S_e$, there exists a corresponding rational place P_α of K_e. The size of S_e is precisely $r^e(r-1)$.

Extending the field of constants. Following [GX22, GRZ21], the actual function field we use is $K_e\mathbb{F}_{q^m} \cong K_e \otimes \mathbb{F}_q^m$, the compositum of F and a finite extension field $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$. The field $K_e\mathbb{F}_{q^m}$ is still defined via the relations (3), except that the field of constants is changed from $\mathbb{F}_q$ to $\mathbb{F}_{q^m}$. Each rational place P of K_e extends to a unique rational place of $K_e\mathbb{F}_{q^m}$, which we continue to denote by P. By linearity, each rational divisor D of K_e extends to a unique rational divisor of $K_e\mathbb{F}_{q^m}$, which we continue to denote by D. The Riemann–Roch space $L(D)$ of K_e and that of $K_e\mathbb{F}_{q^m}$ have the same dimension, and an $\mathbb{F}_q$-basis of the former is also an $\mathbb{F}_{q^m}$-basis of the latter [GX22].

The Frobenius automorphism. The field $K_e\mathbb{F}_{q^m} \cong K_e \otimes \mathbb{F}_q^m$ is equipped with a Frobenius automorphism over $\mathbb{F}_q$, which sends $a \otimes b$ to $a \otimes b^q$. We denote this automorphism by σ. For every place P of K_e, the corresponding place of $K_e\mathbb{F}_{q^m}$ is fixed by σ since K_e is fixed by σ. For $f \in K_e\mathbb{F}_{q^m}$, denote by f^σ the element $\sigma(f)$.

Next, we formally describe the AG code we use.

Definition 1.12. Let $r > 1$ be a prime power and $q = r^2$. Let $n, k, m, e \in \mathbb{N}^+$ be such that $k \le n$ and $n \le r^e(r-1)$. Let $\alpha_1, \ldots, \alpha_n \in S_e \subseteq \mathbb{F}_q^e$ be distinct and let $P_i = P_{\alpha_i}$ for $i \in [n]$. The code $\mathsf{GS}_{q,m,e}(n,k)$ *with evaluation points* $P_1, \ldots, P_n$ is the linear code over $\mathbb{F}_{q^m}$ defined by the following encoding map

$$\begin{aligned} L((k-1)P_\infty) &\to \mathbb{F}_{q^m}^n \\ f &\mapsto C_f := (f(P_1), f(P_2), \ldots, f(P_n)), \end{aligned}$$

where $L((k-1)P_\infty)$ is the Riemann–Roch space of $K_e\mathbb{F}_{q^m}$ associated with the divisor $(k-1)P_\infty$.

The following theorem states that $\mathsf{GS}_{q,m,e}(n,k)$ is an explicit code and provides bounds on its rate and minimum distance.

Theorem 1.13 ([GX22])**.** *The rate of* $\mathsf{GS}_{q,m,e}(n,k)$ *is at least* $(k-g)/n \ge (k-r^e)/n$*, and its minimum distance is at least* $n-k+1$*. Moreover, its encoding map* $f \mapsto C_f$ *can be computed in time* $\mathrm{poly}(r^e, m\log q)$*.*

We also need a map that embeds the message space $L((k-1)P_\infty)$ into $\mathbb{F}_{q^m}^k$.

Definition 1.14. Every $f \in L((k-1)P_\infty)$ has a Laurent series expansion[3] in $T = 1/x_e$ at the place P_∞:

$$f = f_0 T^{-(k-1)} + f_1 T^{-(k-1)+1} + f_2 T^{-(k-1)+2} + \cdots,$$

where $f_i \in \mathbb{F}_{q^m}$. Define $\phi : L((k-1)P_\infty) \to \mathbb{F}_{q^m}^k$ to be the $\mathbb{F}_{q^m}$-linear map that outputs the first k coefficients of this Laurent series, i.e., $\phi(f) = (f_0, f_1, \ldots, f_{k-1})$. The kernel of ϕ is $L(-P_\infty) = 0$. So ϕ is injective.

1.4 List Recoverability of the AG Codes

Recall that the encoding map of $\mathsf{GS}_{q,m,e}(n,k)$ sends a message f to $C_f = (f(P_1), \ldots, f(P_n))$. For subsets $S_1, \ldots, S_n \subseteq \mathbb{F}_{q^m}$, define the agreement $\mathrm{agr}(f, (S_1, \ldots, S_n)) := \{i \in [n] : f(P_i) \in S_i\} \subseteq [n]$.

Next, we prove the following theorem on the list recoverability of the code $\mathsf{GS}_{q,m,e}(n,k)$.

Theorem 1.15. *Let* $s \in [m]$ *and* $t, d \in \mathbb{N}^+$ *such that*

$$(s+1)(d-r^e+1) + k - 1 > n\ell \tag{4}$$

and

$$t > d + k - 1. \tag{5}$$

Identify $\mathbb{F}_{q^m}^k$ *with* $\mathbb{F}_q^{km}$ *by fixing an* $\mathbb{F}_q$*-basis of* $\mathbb{F}_{q^m}$*. Then there exists an affine shift* W *of a* $(k, m, s-1)$*-BTT subspace of* $\mathbb{F}_{q^m}^k$ *such that for any subsets* $S_1, \ldots, S_n \subseteq \mathbb{F}_{q^m}$*, each of size at most* ℓ*, the set*

$$\mathcal{L}_{S_1,\ldots,S_n} := \{f \in L((k-1)P_\infty) : |\mathrm{agr}(f, (S_1, \ldots, S_n))| \ge t\}$$

satisfies $\phi(\mathcal{L}_{S_1,\ldots,S_n}) \subseteq W$*. Moreover, an affine basis of* W *over* $\mathbb{F}_q$ *can be found in time polynomial in* n*,* r^e*, and* $m\log q$*.*

[3]Formally, we have an $\mathbb{F}_{q^m}$-linear field embedding from $K_e\mathbb{F}_{q^m}$ to the ring of formal Laurent series $\mathbb{F}_{q^m}((T)) = \{f_0 T^d + f_1 T^{d+1} + \cdots : d \in \mathbb{Z}\}$, which sends $1/x_e$ to T. The image of $f \in K_e\mathbb{F}_{q^m}$ under this embedding is its Laurent expansion at P_∞.

To prove Theorem 1.15, we need the following claim.

Claim 1.16. Under the notations and conditions of Theorem 1.15, there exists a nonzero polynomial $Q = A_0 + A_1Y_1 + A_2Y_2 + \cdots + A_sY_s$, where $A_0, A_1, \ldots, A_s \in K_e\mathbb{F}_{q^m}$, $A_0 \in L((d+k-1)P_\infty)$, and $A_i \in L(dP_\infty)$ for $i = 1, 2, \ldots, s$, such that for all $f \in \mathcal{L}_{S_1,\ldots,S_n}$,

$$Q(f, f^\sigma, \ldots, f^{\sigma^{s-1}}) = A_0 + A_1 f + A_2 f^\sigma + \cdots + A_s f^{\sigma^{s-1}} = 0. \tag{6}$$

Proof. Choose $Q \neq 0$ such that for all $i = 1, \ldots, n$ and $y_i \in S_i$,

$$A_0(P_i) + A_1(P_i)y_i + A_2(P_i)y_i^\sigma + \cdots + A_s(P_i)y_i^{\sigma^{s-1}} = 0 \tag{7}$$

To see that Q exists, note that A_0 lives in the vector space $L((d+k-1)P_\infty^{(m)})$, which has dimension at least $(d+k-1) - g + 1$ by Riemann's inequality (Theorem 1.10). Similarly, for each $i \in [s]$, A_i lives in $L(dP_\infty^{(m)})$, which has dimension at least $d - g + 1$. This gives

$$\begin{aligned}\dim_{\mathbb{F}_{q^m}} L((d+k-1)P_\infty) + s \cdot \dim_{\mathbb{F}_{q^m}} L(dP_\infty) &\geq (d+k-1) - g + 1 + s(d-g+1) \\ &= (s+1)(d-g+1) + (k-1)\end{aligned}$$

unknowns in total. On the other hand, (7) gives at most $n\ell$ homogeneous linear constraints in these unknowns. By (4), the number of unknowns is greater than the number of linear constraints. So a nonzero solution Q exists.

Consider $f \in \mathcal{L}_{S_1,\ldots,S_n}$ and define $Q^* = A_0 + A_1 f + A_2 f^\sigma + \cdots + A_s f^{\sigma^{s-1}}$. We want to prove that $Q^* = 0$. As $f \in L((k-1)P_\infty)$, $A_0 \in L((d+k-1)P_\infty)$, $A_i \in L(dP_\infty)$ for $i \in [s]$, and σ fixes P_∞, we know $Q^* \in L((d+k-1)P)$.

Consider $i \in [n]$ such that $f(P_i) = y_i$ for some $y_i \in S_i$. By (7), we have

$$\begin{aligned}0 &= A_0(P_i) + A_1(P_i)y_i + A_2(P_i)y_i^\sigma + \cdots + A_s(P_i)y_i^{\sigma^{s-1}} \\ &= A_0(P_i) + A_1(P_i)f(P_i) + A_2(P_i)(f(P_i))^\sigma + \cdots + A_s(P_i)(f(P_i))^{\sigma^{s-1}} \\ &= A_0(P_i) + A_1(P_i)f(P_i) + A_2(P_i)f^\sigma(P_i) + \cdots + A_s(P_i)f^{\sigma^{s-1}}(P_i) \\ &= (A_0 + A_1 f + A_2 f^\sigma + \cdots + A_s f^{\sigma^{s-1}})(P_i) \\ &= Q^*(P_i),\end{aligned}$$

where the third equality uses the fact that $(f(P_i))^\sigma = f^\sigma(P_i)$, which holds since P_i is fixed by σ.

As $|\mathrm{agr}(f, (S_1, \ldots, S_n))| \geq t$, the above argument shows that there exist $i_1, \ldots, i_t \in [n]$ such that Q^* vanishes at $P_{i_1}, \ldots, P_{i_t}$. Let $D = \sum_{j=1}^t P_{i_j}$. Then $Q^* \in L((d+k-1)P_\infty - D)$. The degree of the divisor $(d+k-1)P_\infty - D$ is $d+k-1-t$, which is less than zero by (5). So $L((d+k-1)P_\infty - D) = 0$, implying that $Q^* = 0$. □

Proof sketch of Theorem 1.15. The rest of the proof of Theorem 1.15 is identical to the proof in [GRZ21], which extends a proof in [GX22]. We sketch the main idea: Expand $A_0, A_1, \ldots, A_s$ and f into Laurent series in $T = 1/x_e$ at the place P_∞. Substitute these into (6) and examine the coefficients of $T^{-(k-1)+i}$ for $i = 0, 1, \ldots, k-1$. This yields k linear equations over $\mathbb{F}_{q^m}$. Identifying $\mathbb{F}_{q^m}$ with $\mathbb{F}_q^m$, we obtain km linear equations over $\mathbb{F}_q$. By analyzing the structure of these linear equations, one finds that their homogeneous parts form a (k, r, m)-BTT matrix M for some $r \geq m - s + 1$. So $\phi(f)$ is in an affine shift of the kernel of M. Finally, by [GRZ21, Claim 3], the kernel of M is contained in the image of a linear map represented by a $(k, m, s-1)$-BTT matrix, which by definition is a $(k, m, s-1)$-BTT subspace. □

1.5 Putting It Together

We now combine the existence of explicit BTT-evasive subspaces (Theorem 1.6) with Theorem 1.15 to prove Theorem 1.1.

Theorem 1.17 (Restatement of Theorem 1.1)**.** *Let $\varepsilon > 0$ and $\ell \in \mathbb{N}^+$. Let $q \geq (4/\varepsilon + 1)^2$ be an even power. Choose $R \in (0,1)$. There is some $m_{\min} = O(\ell \log_q(\ell/\varepsilon)/\varepsilon^2)$ such that the following holds. For all large enough n of the form $q^{e/2}(\sqrt{q}-1)$, where $e > 0$ is an integer, and all $m \geq m_{\min}$, there is a deterministic construction of an $\mathbb{F}_q$-linear code $C \subseteq \mathbb{F}_{q^m}^n$ over the alphabet $\mathbb{F}_{q^m}$ of rate at least R and relative distance $1-R-O(\varepsilon)$ that is $(1-R-\varepsilon, \ell, L)$ list-recoverable in time $\mathrm{poly}(L,n)$, returning a list of size at most $L := \exp_q(\mathrm{poly}(\ell/\varepsilon))$ that is contained in a subspace over $\mathbb{F}_q$ of dimension at most $\mathrm{poly}(\ell/\varepsilon)$.*

Proof. We may assume $1-R-\varepsilon \geq 0$, i.e., $R \leq 1-\varepsilon$. Let $e \in \mathbb{N}^+$ be a growing parameter. Choose the following parameters:

- Let $\varepsilon' = \varepsilon/4$.
- Let $r = \sqrt{q} \geq 1/\varepsilon' + 1$.
- Choose $n = q^{e/2}(\sqrt{q}-1) = r^e(r-1) \geq r^e/\varepsilon'$.
- Let $k = \lceil (R+2\varepsilon')n \rceil \leq n$. By choosing e to be large enough, we may assume $k \geq k_{\min}$ where $k_{\min} = \exp(\mathrm{poly}(r/\varepsilon))$ as in Theorem 1.6.
- Let $s = \lceil \ell/\varepsilon' \rceil$. Then $\ell \leq \varepsilon' s$.
- Choose large enough $m_{\min} = \Theta(\ell \log_q(\ell/\varepsilon)/\varepsilon^2)$, and let $m \geq m_{\min}$.

Consider the code $\mathsf{GS}_{q,m,e}(n,k)$, which is a linear code over $\mathbb{F}_{q^m}$. By Theorem 1.13, the rate of $\mathsf{GS}_{q,m,e}(n,k)$ is at least

$$(k - r^e)/n \geq R + 2\varepsilon' - r^e/n \geq R + \varepsilon',$$

and its relative distance is at least $(n-k+1)/m = 1 - R - O(\varepsilon)$.

Let $d = \frac{n\ell - k + 2}{s+1} + r^e - 1$ so that (4) is satisfied, and let $t = d + k$ so that (5) is satisfied. Then with this setting of parameters, we know from Theorem 1.15 that $\mathsf{GS}_{q,m,e}(n,k)$ can be list recovered up to the radius

$$\begin{aligned} n - t = n - d - k &= n - k + 1 - g - \frac{n\ell - k + 2}{s+1} \\ &\geq n - (R+2\varepsilon')n - r^e - \frac{n\ell}{s+1} \\ &\geq n - Rn - 4\varepsilon' n \\ &= (1 - R - \varepsilon)n. \end{aligned}$$

Moreover, the image of the list of candidate messages under ϕ is contained in an affine shift of a $(k, m, s-1)$-BTT subspace over $\mathbb{F}_q$, where $s - 1 \leq \ell/\varepsilon' \leq \varepsilon' m$.

Finally, define the final code C to be $\mathsf{GS}_{q,m,e}(n,k) \cap \phi^{-1}(W)$, where $W \subseteq \mathbb{F}_q^{km}$ is an explicit $(k, m, s-1, A)$-BTT evasive subspace of codimension at most $\varepsilon' km$ and $A = \mathrm{poly}(s/\varepsilon') = \mathrm{poly}(\ell/\varepsilon)$. The subspace W can be explicitly constructed by Theorem 1.6. As $\phi^{-1}|_{\mathsf{GS}_{q,m,e}(n,k)}$ is injective, intersecting $\mathsf{GS}_{q,m,e}(n,k)$ with $\phi^{-1}(W)$ reduces its rate by at most ε'. So the rate of C is at least $(R+\varepsilon') - \varepsilon' = R$. Moreover, intersecting $\mathsf{GS}_{q,m,e}(n,k)$ with $\phi^{-1}(W)$ reduces the $\mathbb{F}_q$-dimension of the affine space containing the list of candidate messages to $A = \mathrm{poly}(\ell/\varepsilon)$. The list size is at most $q^A = \exp_q(\mathrm{poly}(\ell/\varepsilon))$. □